

VINAY K R

Senior GenAI & Applied AI Systems Engineer

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PROFESSIONAL SUMMARY

Senior GenAI & Applied AI Systems Engineer with **3+ years of enterprise experience** architecting high-throughput distributed microservices, autonomous agentic workflows, and production Generative AI platforms across fintech (**Liminal Custody**) and gaming technologies (**Light & Wonder**). Specialized in **Advanced Hybrid RAG (pgvector HNSW + BM25 + Reciprocal Rank Fusion)**, **Multi-Agent Orchestration (LangGraph, Semantic Kernel, Model Context Protocol [MCP])**, and **Local LLM Serving (vLLM PagedAttention, Redis semantic caching, LoRA/QLoRA 4-bit fine-tuning)**. Proven track record bridging state-of-the-art AI models with resilient backend infrastructure (Python FastAPI, .NET 10 LTS Minimal APIs, Go, Angular 22 Signals). Creator of an independent **21-service microservices platform load-tested at 500 RPS (p95 <85ms)**; authored **750+ automated E2E test suites** with zero production compliance regressions.

TECHNICAL EXPERTISE

GenAI & Agents:	LangGraph (Cyclic Graphs, Reflection), Semantic Kernel 1.79, Model Context Protocol (MCP), ReAct Agent Loops, Tool Calling, PydanticAI, NeMo Guardrails, Guardrails AI
RAG & Vector Stores:	Advanced Modular RAG, PostgreSQL 18 + pgvector (HNSW Cosine), Qdrant, BM25 Lexical Search, Reciprocal Rank Fusion (RRF, k=60), Contextual Chunking, Cross-Encoder Rerankers
Inference & Tuning:	vLLM (PagedAttention, Continuous Batching), Ollama, 4-bit QLoRA/LoRA (Unsloth, PEFT), Redis 8 Vector Semantic Caching (<5ms latency), SSE Token Streaming
Evals & Observability:	Ragas 0.2 (Faithfulness, Answer Relevance, Context Recall), DeepEval, LLM-as-a-Judge, OpenTelemetry, Arize Phoenix, Jaeger Distributed Tracing
Backend & Cloud:	Python 3.12+ (FastAPI, Pydantic v2, Asyncio), C# 14 (.NET 10 LTS Minimal APIs), Go (chi), PostgreSQL 18, Redis 8.10, Docker, Kubernetes (k3s), Argo CD GitOps, Azure AI / Key Vault
Frontend & Testing:	Angular 22 (Signals, linkedSignal, resource, Zoneless CD), TypeScript 7, Cypress (750+ E2E Tests), Pytest-asyncio, xUnit v3

PROFESSIONAL EXPERIENCE

Software Engineer – Full Stack & AI Systems | *Liminal Custody* Nov 2025 – Mar 2026 | Bengaluru, India

- Architected and shipped the core **Transaction Firewall Policy Engine** and AI risk-scoring rules engine, enabling real-time threat detection and crypto asset transfer governance across 5 enterprise screen families.
- Designed rule-condition-action evaluation pipelines with **Redis 8 vector and memory caching**, reducing rule evaluation latency by **65%** and integrating TRM Labs threat intelligence for real-time cryptocurrency address risk scoring.
- Engineered multi-organization RBAC and claim-scoping middleware with JWT authorization, Angular route guards, and HTTP interceptors, securing multi-tenant digital asset custody operations.
- Implemented multi-signature quorum consensus validation for policy mutations, enforcing enterprise threshold controls (e.g., \$100K/24hr window) prior to critical transaction execution.

Senior Associate Software Engineer | *Light & Wonder (LNW)* Aug 2023 – Jul 2025 | Bengaluru, India

- Modernized 20 enterprise modules and 50+ UI screens across 4 casino product lines (Cage Credit, Servizio, Engage telemetry, iView displays) interfacing with high-throughput C#/.NET Core Web APIs.
- Architected a **Game Recommendation Engine** utilizing semantic clustering and real-time player telemetry, increasing user engagement by **18%**.
- Eliminated critical video-memory leak causing slot-machine crashes during long-running media playback by engineering media caching and ngOnDestroy lifecycle teardowns.
- Authored and maintained **750+ Cypress automated E2E regression test suites** across Servizio (~300 tests) and Engage with one peer, sustaining zero-defect compliance through regulated gaming certification releases.
- Built real-time SQL Server CDC and trigger audit telemetry across 100+ database tables with custom reporting interfaces for strict regulatory compliance audits.

Full Stack Intern | *Light & Wonder (LNW)* Mar 2023 – Jul 2023 | Bengaluru, India

- Developed a Game Recommendation System using C#/.NET Core Minimal APIs and Angular, mastering asynchronous programming, database query optimization, and REST API design.

PRODUCTION AI & SYSTEMS PROJECTS

Enterprise Agentic RAG Platform

Python FastAPI · LangGraph · pgvector · MCP · Angular 22 | 2026

- Designed an enterprise Multi-Agent RAG platform supporting multi-format document ingestion (.pdf, .docx, .xlsx, .md) with **pgvector HNSW indexing** and citation-grounded SSE streaming.
- Implemented **Hybrid Retrieval** fusing dense embeddings with PostgreSQL BM25 via **Reciprocal Rank Fusion (RRF, k=60)**, improving domain retrieval recall by **34%**.
- Integrated **Model Context Protocol (MCP)** tool execution, enabling AI agents to autonomously query enterprise databases and external compliance APIs.
- Built continuous evaluation pipelines using **Ragas** (Faithfulness > 0.92, Context Recall > 0.88) with end-to-end OpenTelemetry distributed tracing.

High-Throughput Local LLM Inference Gateway & LoRA Pipeline

FastAPI · vLLM · Redis · Unsloth · PEFT | 2026

- Built an asynchronous Python FastAPI inference gateway with **Redis 8 vector semantic caching**, returning **sub-5ms cached responses** for 40%+ of repetitive queries.
- Deployed quantized 8B local models via **vLLM with PagedAttention and continuous batching**, slashing cloud LLM API costs by **60%**.
- Engineered an automated **4-bit QLoRA fine-tuning pipeline** using Unsloth and PEFT on specialized domain instruction datasets with NeMo Guardrails for input sanitization and prompt injection defense.

Medha — Distributed Microservices Platform (Scale Proof)

Go · PostgreSQL/PostGIS · Redis · Docker · k3s · Argo CD | Aug 2025 – Present

- Architected an independent distributed backend with **21 bounded microservices**, **~200 REST endpoints**, 50 database migrations, and 109 automated integration tests.
- Validated high-concurrency throughput under distributed load testing at **500 RPS with 100% success rate and p95 latency <85ms**.
- Implemented PostGIS proximity search using `ST_DWithin` over GIST spatial indexes with distance-ordered keyset pagination.
- Deployed via **Argo CD GitOps on Kubernetes (k3s)** with dev/prod namespace isolation, default-deny NetworkPolicies, SealedSecrets, and Cloudflare Tunnels.

OPEN SOURCE & AI SYSTEMS ENGINEERING CONTRIBUTIONS

FastAPI & LangGraph Production Agent Patterns

Python · LangGraph · OpenTelemetry | 2026

- Authored production-ready reference patterns for stateful LangGraph multi-agent supervisors with human-in-the-loop checkpoints and async token emitters.
- Engineered a high-performance Redis 8 vector semantic caching wrapper achieving sub-5ms exact match evictions for recurring LLM prompt patterns.

LEADERSHIP, COMPLIANCE & ENGINEERING RIGOR

Enterprise Test Automation & Gaming Certification

Light & Wonder · Liminal Custody | 2023 – 2026

- Spearheaded E2E test automation across 2 critical casino platform product lines, maintaining 750+ Cypress tests and achieving zero compliance regressions across state gaming lab audits.
- Standardized pull request review gates, database migration auditing (goose), and secure secret management with SealedSecrets across Kubernetes deployments.

EDUCATION & ACADEMIC CREDENTIALS

Siddaganga Institute of Technology (SIT), Tumakuru, Karnataka, India

Aug 2019 – Jul 2023

Bachelor of Engineering (B.E.) in Computer Science & Engineering

CGPA: 8.65 / 10.0